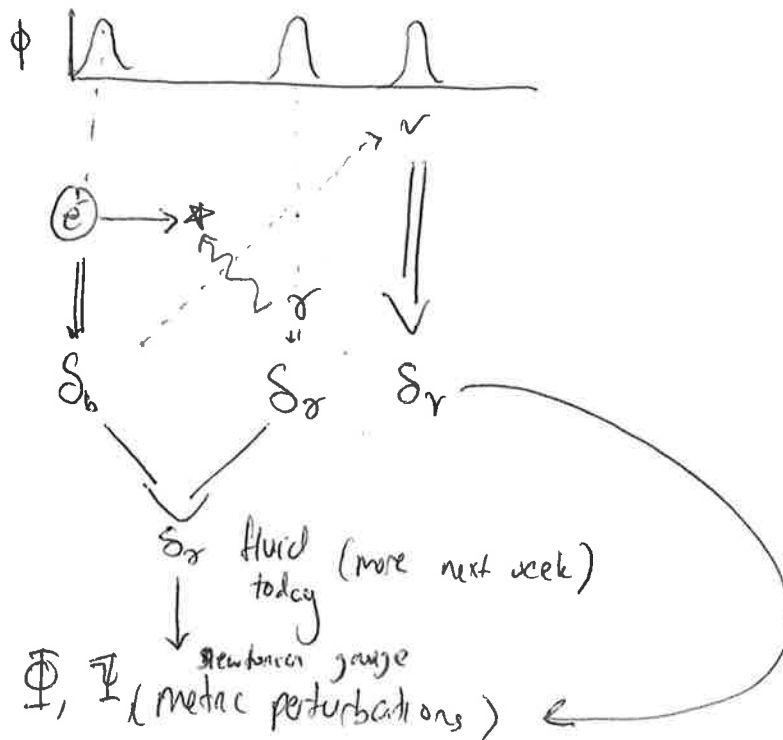


Today: start perturbations!



Whole system messy. Today simple: δ Fluid + ν

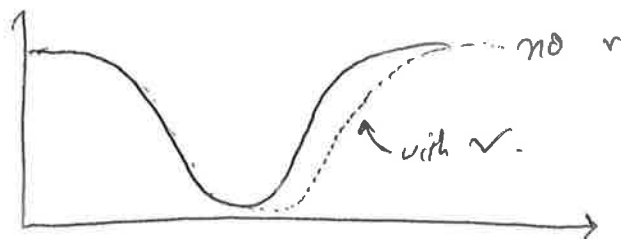
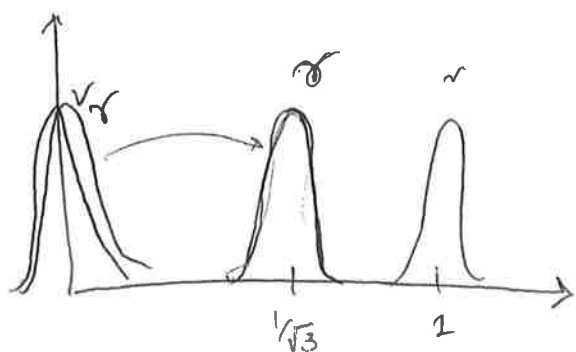
Cute physics: phase shift

Motivation

→ We want unique sig. of some physics we care about.

→ change ~~any~~ most things (Ω_b , Ω_c , Ω_Λ , DM interaction, inflation) signature can be mimicked by something else.

→ however ν ~~then~~ (and e.g. quarkonium with $v/c \ll 1$) have a very unique signature



the "grow" influence of v pulls γ out.

$$\Rightarrow \delta_\gamma(x; \text{no } v) = \delta_\gamma(x + \delta x; v)$$

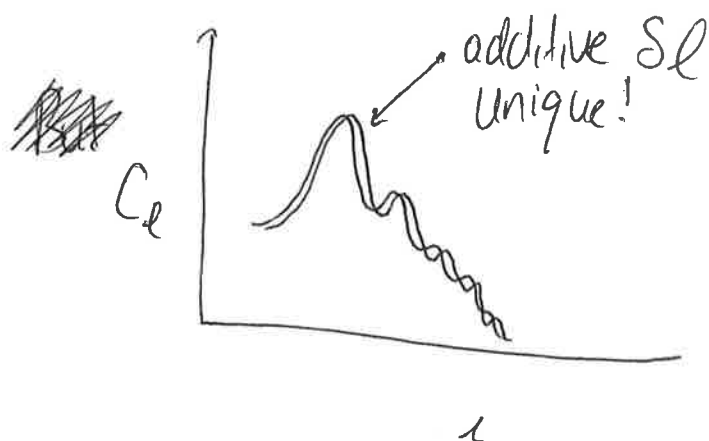
→ true for any species that travels faster than $c \sim \frac{1}{\sqrt{3}}$.

displacement in real space $\Leftrightarrow$ phase shift

$$\mathcal{F}[f(x + \delta x)] \sim \int e^{ikx} f(x + \delta x) = \int e^{ik(y - \delta x)} f(y)$$

$$= e^{-ik\delta x} \int e^{iky} f(y)$$

$$= e^{-ik\delta x} \mathcal{F}[f(x)]$$



(claim by some e.g. Kamen
that phase shift $\Leftrightarrow$ Net,
probably not true, e.g. look at
 $\frac{\Delta C_d}{\sqrt{C_d}} \sim \cos \alpha$ or
phase shift basically nothing)

What ~~the~~ eq. we need to numerically evolve?

→ don't want too much abt. explicit form, you're on night of mathematics away from deriving them. Let's analyze instead.

$$d\sigma, d\tau, u_a \rightarrow \vec{v}_a = -\vec{\nabla} u_a \quad ds^2 = a^2 [-(1+2\Phi)d\tau^2 + (1-2\Psi)d\vec{r}^2]$$

σ perfect fluid (σ -e tight coupled)

$$\rightarrow d\sigma' = -k^2 u_\sigma \approx \delta\sigma \sim -\nabla \cdot \vec{v}_\sigma \quad \text{continuity.}$$

" $F = ma$ "

$$u_\sigma' = \frac{1}{3} d\sigma + (\Phi + \Psi)$$

" a "

" F " $\sim$ pressure + grav
 ↓
 more of smeth.
 something $\rightarrow$ more pressure $\sim$ energy density.
 $\Phi + \Psi$ metric potential.

$$\rightarrow d_r' = -k^2 u_r$$

new!

$$u_v' = \frac{1}{3} d_r + \underbrace{\Phi + \Psi}_{\text{"energy density"}} - k^2 \underbrace{\pi_v}_{\text{new!}}$$

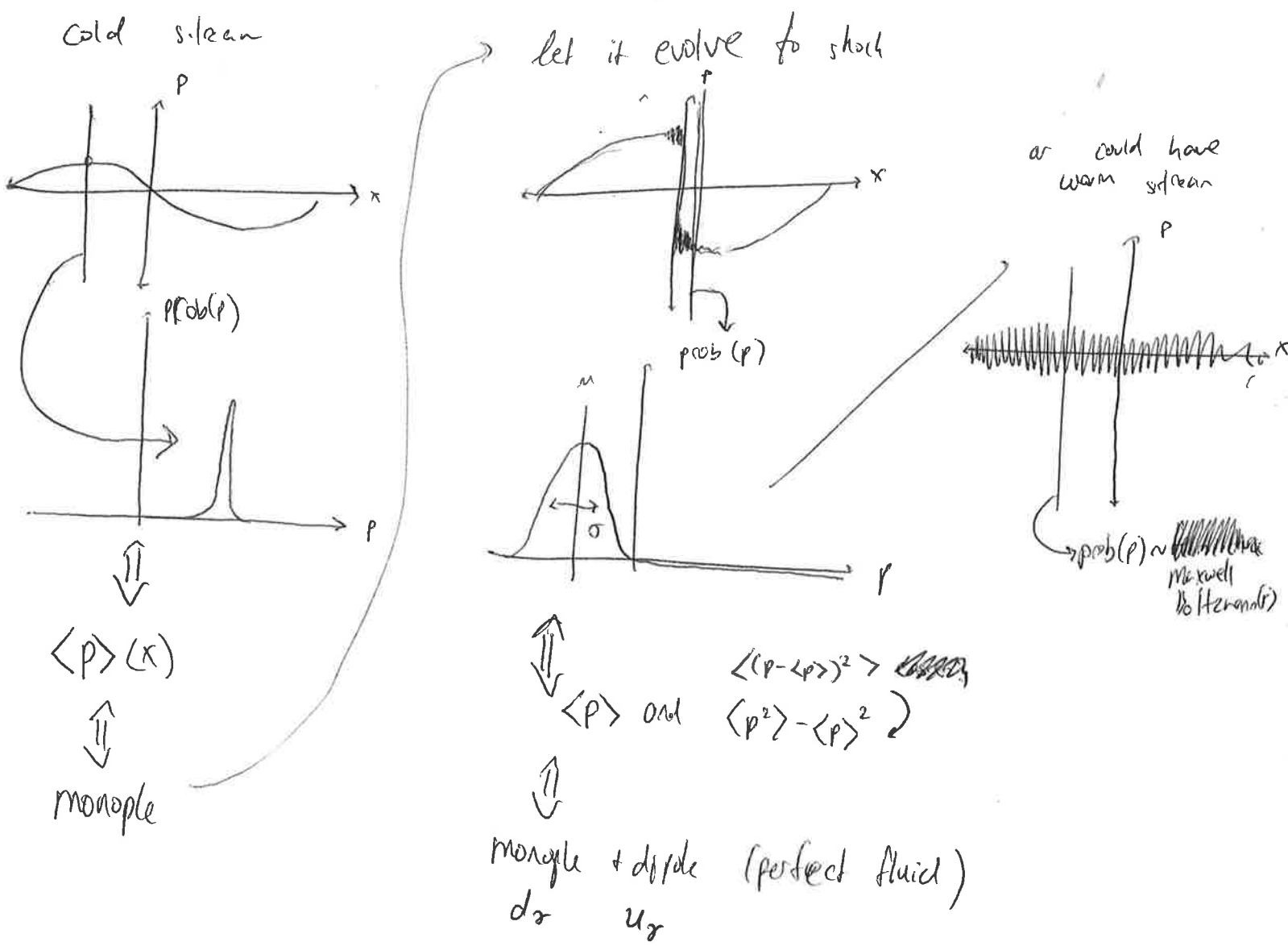
Pressure deficit radiat.
of $\nu\nu$ scattering.

what the heck is this?

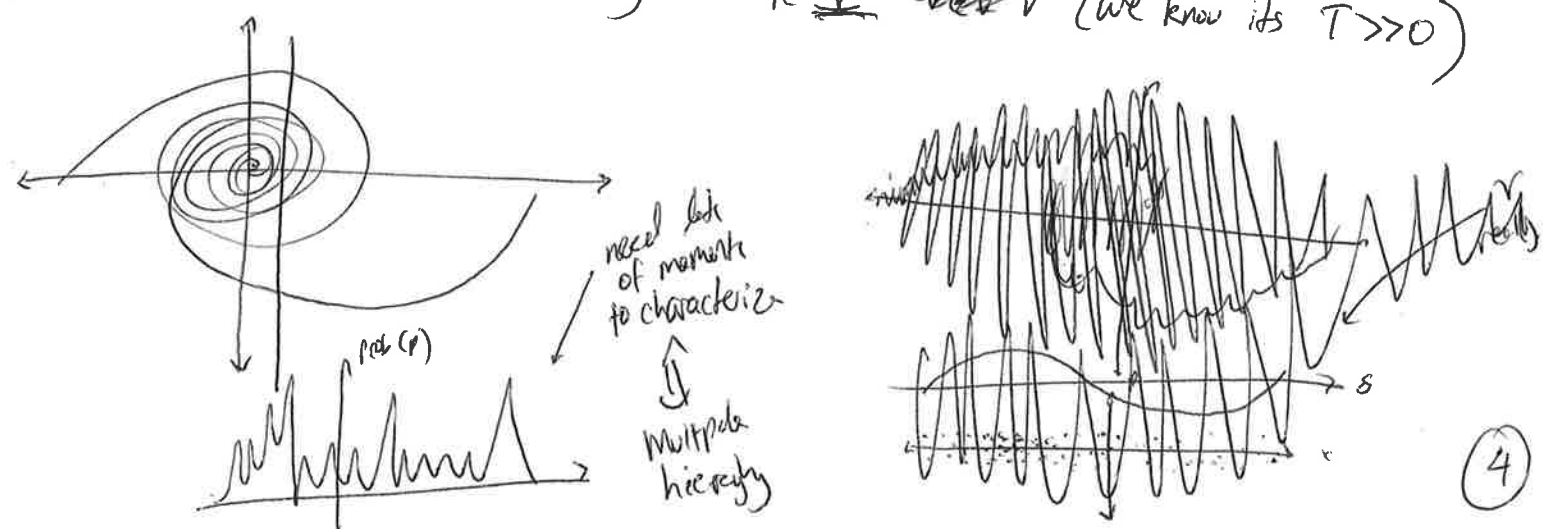
multiples $\rightarrow$ some intuition $\rightarrow$

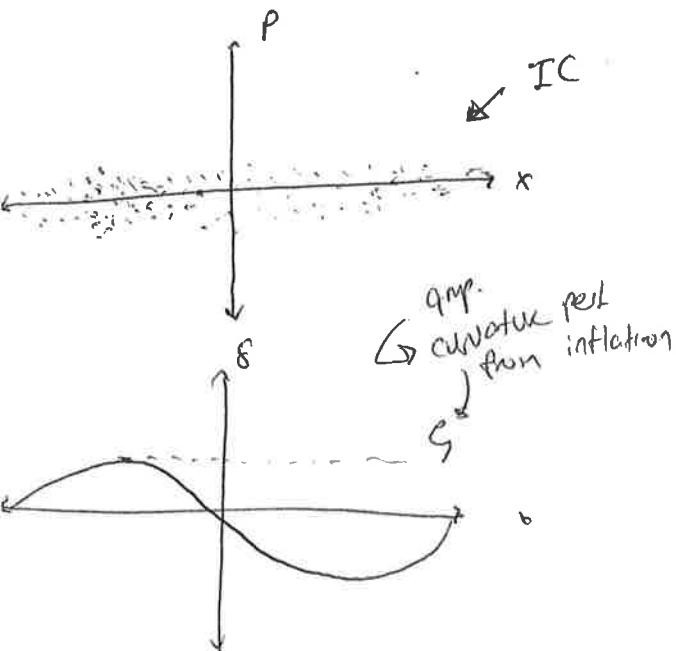
monopole formal def: angle average exp with Legendre polynomials. yuck.

but intuition ~~the~~ is the following



but what if collisions? (e.g. DM during NL collapse ~~or~~ ~~at~~ V (we know its $T \gg 0$))





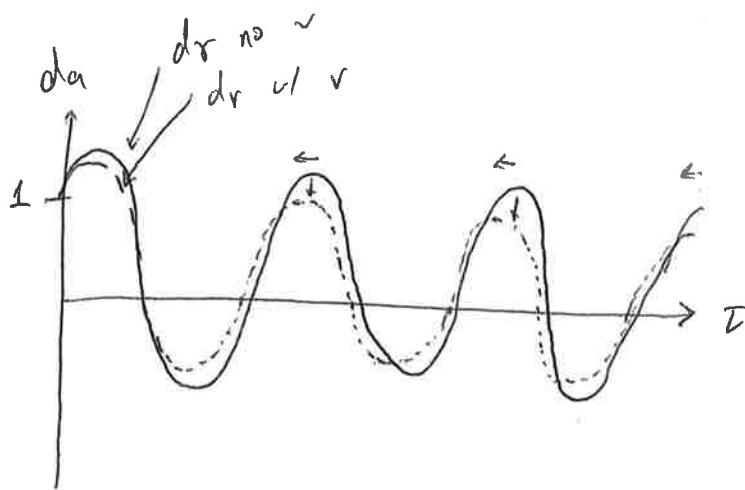
Quiz: ~~do~~ do you need

e.g. ~~quadruple~~ in VIC?
 $l=3$ hexadecapole

Answer: NO. (anything above ~~quadruple~~ $\rightarrow 0$)

anyways. Rq you need in handout.

what you will compute.



phase shift
 +
 amplitude modulation

$\rightarrow$ you can also compute analytically ~~what~~ $\mathcal{E}_2$ SP and $(1 + \Delta \delta)$
 BS03, matches very well.